Self-Optimizing Power Grid: AI for Real-Time Energy Fraud Detection & Load Balancing

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Abstract- *The increasing power demand in urban environments requires an efficient, sustainable and reliable power system. Modern power grids have become vulnerable to inefficiencies in load balance and prone to malpractices like power theft. Conventional rule-based and manual methods fail to deliver reliability, security and efficiency under dynamic conditions creating the need for an AI driven framework. This report proposes a self-optimizing power grid framework that leverages artificial intelligence for real time fraud detection and load balancing. This work integrates probabilistic fraud scores from Vertex AI logistic regression directly into PPO-based load balancing states to enable fraud-aware optimization, reducing transformer overloads by 35-45% in Mumbai-like synthetic scenarios. Due to privacy restrictions and the rarity of labeled fraud data, Gaussian-based synthetic datasets were generated to preserve realistic temporal, environmental ,and inter-feature correlations characteristic of Mumbai's urban power grid. For grid management, we applied a Proximal Policy Optimization (PPO) agent that learns how to distribute loads efficiently, while respecting transformer safety limits, maintaining frequency stability, and adapting to patterns in time and renewable energy contribution. Our results show that the approach not only improves fraud detection-achieving an accuracy of 79.91%, but also reduces transformer overload events by 35-45% and mitigates non-technical losses through fraud-aware adjustments preserving stability and security in a dynamic operating environment. The proposed framework aligns with SDG Goal 7 (Affordable and Clean Energy) and Goal 11 (Sustainable Cities and Communities) by fostering a resilient, secure, and sustainable power infrastructure. The source code, including data generation, model training, and Flask interface.*

Keywords- *Smart Grid; Big Data Analytics; Reinforcement Learning; Fraud Detection; Synthetic Data Generation; Urban Energy Systems; Proximal Policy Optimization.*

1. INTRODUCTION

Urban power distribution systems in large metropolitan cities under growing pressure of population expansions, denser cities and electricity consumption patterns have unprecedented pressure. These trends lead to generation of large scale , high velocity smart meter data . Mumbai being India's most populous city and a major economic centre demands modern power grid catering to the consequent fluctuations intensified by seasonal changes, especially during extreme summers and monsoon season. Electricity distribution and transmission losses are categorised as technical and non-technical losses (NTL). Due to inefficiencies in the equipment and infrastructure technical losses occur while NTL are the result of billing errors,electricity theft etc.NTL is a global concern which costs utilities about 96 USD billion annually [1,2].

In urban areas like Mumbai, where demand instability and electricity theft prevail, then data reliability becomes significant for effective grid control. Load balancing decisions that are made using unverified consumption data may amplify the effects of fraud, while excessive aggressive control actions can mask abnormal usage patterns. The incompetency in detecting complex fraudulent patterns due to the conventional rule-based and manual inspection methods are also inefficient [3,4]. Recent work employs traditional classifiers like Gradient Boosting and Random Forest to advanced deep learning architectures like Gated Recurrent Units (GRUs) and Convolutional Neural Networks (CNNs) [1,5,6]. Meanwhile, reinforcement learning (RL) is adopted for grid optimization since load scheduling can be formulated as a Markov Decision Process (MDP) allowing RL algorithms such as Q-Learning and Deep Q-Networks to dynamically balance supply and demand [7]. Despite this progress a critical issue persists that is the limited availability of real world, labeled fraud data due to privacy concerns and its rarity of verified fraud incidents hampering the development of reliable models.

This study presents a Self-Optimizing, big-data driven power grid framework that unites machine learning based fraud detection with reinforcement learning driven load balancing using a closed loop system. Smart meter data for fraud detection are stored

and processed using BigQuery Big tables, enabling scalable ingestion and analytics, while Vertex AI used for cloud-native model training and deployment. We get probabilistic fraud outputs and use Proximal Policy Optimization (PPO) based load balancing agent, enabling load balancing.Developed with scalability and real-time deployment in mind, enhances grid stability and reduces transformer stress.

1.1 Organisation of the paper:

1. The primary focus of this research is to comprehensively examine an ai driven self-optimizing smart grid framework that integrates real time electricity for detection with dynamic load balancing targeting the challenges faced by the dense urban power grid such as Mumbai.
2. The paper discusses the background and motivation behind the proposed framework highlighting the limitations of traditional creative monitoring systems and growing impact of electricity theft in urban environments.
3. In-depth review of existing literature on smart grid fraud detection using BigQuery- based data storage, Vertex AI and rural based load balancing using PPO is provided emphasizing performance metrics ,current techniques and unresolved challenges in standalone approaches.
4. The study formulates research problems and presents proposed system architecture describing the end-to-end workflow which includes data ingestion, fraud detection and fraud aware state construction and load optimization.
5. The paper presents experimental results and performance analysis evaluating the effectiveness of the proposed framework in reducing overload events improving demand supply balance and handling fraudulent consumption scenarios
6. The study also discusses limitations, identifies key research caps, suggests future research directions and concludes with a summary of the contributions and practical implications of the proposed system.

1.2 Background Study

a) Mumbai's Power Grid: System Under Constant Pressure

Mumbai's electricity grid is more than cables and transformers, from dense residential high-rises and commercial districts to railway systems, financial institutions and hospitals, electricity demand is tightly packed into a very small geographic area. This extreme pressure creates a highly connected and sensitive distribution network, where stress in one part of the grid can quickly work on others [8, 9]. Unlike cities with balanced land usage , Mumbai's power demand changes significantly across time and location. Intense summer cooling loads, morning and evening household peaks and continuous commercial usage create constant strain on transformers [17,19]. Research on urban smart grids shows that rule-based or static methods find it difficult in such environments because they cannot adjust to real-time changes and consumption behavior [9,25].

b) Reliability Challenges and the Reality of Blackouts

Grid reliability is generally measured using indices such as SAIDI ,which track the average time of power interruptions experienced by consumers. Even in advanced power systems, yearly outage durations remain significant [8,10]. Mumbai's October 2020 blackout exposed the weakness of highly interconnected urban grids. Investigations revealed that grid variations and protection system responses triggered such failures, resulting in a city-wide outage affecting transportation, healthcare services and financial operations [11,12]. Studies on blackout dynamics show that such events often start with localized overloads or inaccurate system states that remain undetected until system limits are surpassed [13].

c) Electricity Theft as a Hidden Stress on the Grid

Meter tampering, billing manipulation and unauthorized connections contribute to significant general losses, directly impacting grid strength and utility revenues [14]. Reports indicate a continuous increase in power theft cases across Mumbai, particularly in heavily populated residential and mixed-use neighborhoods [16]. Electricity theft has affected grid in two critical ways. First,

it bends consumption data, leading to inaccurate demand forecasting and inefficient grid arrangement [14, 15]. Second, illegal utilization often clusters locally, causing transformers to operate beyond their thermal limits without caution. Old detection methods rely on post-billing audits, making them unsuitable for real-time grid protection and control [15].

d) **Demand Changes Driven by Urban Life and Climate**

Mumbai's electricity demand is molded by a complex interaction of human activity, climate, and economic behavior. Residential demand tops during mornings and evenings, while commercial and industrial loads dominate during the day. Seasonal variations further intensify these changes, in summer heatwaves drive heavy air conditioning usage, while monsoon conditions change consumption patterns across the city [17], [19]. Research on load prediction highlights that such time-varying, non-linear relationships are difficult to capture using traditional statistical or rule-based techniques. Without adaptive learning mechanisms, grid operators wrestle to respond effectively to sudden demand surges, increasing the risk of inefficient utilization and transformer overloads[18]. Old age grid management techniques were designed for relatively predictable and centralized power systems. The rule-based control systems assume firm operating conditions and fail when confronted with unexpected shifts[20]. General reinforcement learning methods, such as value-based or actor–critical algorithms, face challenges related to training instability and scalability when applied to real world power systems.

e) **Artificial Intelligence as a Transformative Enabler**

Artificial intelligence offers a very different approach to grid management. Instead of relying on predefined rules, AI systems learn directly from historical and real-time data and authorize continuous adaptation to changing grid conditions. Machine learning models can identify small problems in consumption patterns indicative of electricity theft, while reinforcement learning starts sequential decision-making for dynamic load balancing [15,23]. In particular, policy gradient methods such as Proximal Policy Optimization (PPO) have gained attention for their stability and suitability in controlling problems of power grid operations. Surveys on smart grid modernization emphasize that AI allows grids to transition from reactive operation to predictive and self-correcting systems that improve over time through feedback and learning [22,26]. A self-optimizing grid must integrate real-time fraud detection to ensure data integrity, adaptive load balancing to protect transformers, and continuous learning mechanisms that evolve with changing urban conditions [23,24].

2. LITERATURE REVIEW

This section reviews the two foundational studies that form the basis of the present research. These works represent recent and relevant advances in electricity fraud detection and reinforcement learning based load balancing within smart grid systems.In the Table 1 describes the literature survey that includes the domain, model applied, datasets used and performance metrics in their work.

2.1 Electricity Theft Detection Using Machine Learning

[1] presented a comprehensive study on electricity theft detection in smart grids using machine learning. The work focuses on identifying fraudulent electricity consumption by analyzing smart meter data, which typically exhibits high dimensionality, missing values, and severe class imbalance. The authors employed a combination of traditional machine learning models (such as Random Forest and Gradient Boosting) and deep learning architectures, including a hybrid GRU–MLP model, to capture both short-term and long-term consumption patterns. Time-domain and frequency-domain features were extracted from daily smart meter readings, enabling the model to learn subtle irregularities indicative of electricity theft. Synthetic data generation techniques like k-means SMOTE were employed to handle the imbalance between normal and fraudulent cases. The study demonstrated strong performance across multiple evaluation metrics which includes precision, recall,accuracy F1-score, PR-AUC, and ROC-AUC. These results highlight that machine learning models are efficacious in identifying electricity theft under controlled conditions.

The authors also recognized several limitations. The model training and evaluation were conducted in an offline enviornment with the usage of historical data where fraud detection served as a classification task intended to aid billing audits and inspection processes. The study did not explore how fraud detection outcomes could influence real-time grid operations or how fraudulent consumption might affect load balancing and grid stability.

2.2 Reinforcement Learning for Load Scheduling in Power Grids

Luo [7] explored reinforcement learning based solutions for load scheduling as Markov Decision Processes. Load balancing was formulated as a decision-making problem being sequential, aiming demand supply operation to be optimized along with minimizing costs. The grid environment was modeled as a Markov Decision Process (MDP), with the state space that consists of the demand levels, generation capacity and energy storage states. Multiple reinforcement learning algorithms were analyzed, including Actor–Critic methods,Deep Q-Networks (DQN), and Q-Learning.Based on mean cost reduction, rate of convergence,balance between supply and demand and stability under dynamic load conditions the performance was monitored. The results demonstrated that RL agents can adapt to learn about the load scheduling strategies without relying on analytical models of grid dynamics making it suitable for modern power grids when characterized by behaviours like uncertainty, time variance and non linearity.

Even with these advantages, the study relied on input data to be accurate and reliable. The prevalence of electricity theft, manipulation of data, or anomalous consumption patterns on the learning and decision-making process was not supported. As a result, the proposed framework solution may perform inadequately when deployed in real-world urban grids where non-technical losses are common.

In urban power systems such as Mumbai's, treating these processes separately will possess more challenges. Consumption of fraudulent usage of electricity directly distorts demand data, which serves as a primary input to load balancing algorithms. If load scheduling decisions are based on such data, they can lead to overloading of transformers,poor electricity distribution, and higher risk of outages.

Our study aims to bridge the gap between fraud detection and load balancing using these two foundational studies:

1) Treats fraud detection as a real-time control signal, not just a reporting mechanism
2) Incorporates fraud probability into the reinforcement learning state space
3) Enables fraud-aware load balancing, improving robustness in urban grids

By integrating the strengths of machine learning–based fraud detection [1] with reinforcement learning–based load scheduling [7] the proposed framework advances toward a more realistic, adaptive, and deployable smart grid solution for metropolitan regions.

Table No. 1: Literature Survey of the previous studies.

Year	Domain	Algorithm	Dataset	Performance Metric
[1] 2024	Smart grid fraud detection (electricity theft)	Traditional ML + Deep Learning (Random Forest, Gradient Boosting, MLP-GRU)	China National Grid Corporation (CNGC) dataset: 42,372 consumer records (38,752 honest + 3,615 theft).	Accuracy,Recall, Precision, ROC-AUC,F1-score
[2] 2023	Electricity fraud detection in Smart Metering Systems	Supervised: Wide/Deep **FFNN**, **CNN**; Unsupervised: Clustering (CFSFDP),RNN	Various public and proprietary smart meter datasets (includes SGCC & Irish CER)	Accuracy, robustness against adversarial attacks, privacy preservation
[3] 2022	Fraud Detection (energy)	Supervised (DT, SVM, ANN), Unsupervised (SOM, Clustering)	This study compares various other datasets and doesn't include its own.	Accuracy (Acc), Detection Rate (DR), False Positives (FP)
[4] 2021	Fraud detection(energy)	NGBoost (Natural Gradient Boosting)	Sensors 21, no. 24 paper: dataset not clearly described in abstract; need full text	Accuracy: 93%, Recall: 91%, Precision: 95%, AUC: 0.96
[5] 2022	Electricity fraud detection in Smart Grid	CNN-GRU hybrid tuned with Cuckoo Search (CS)	Forecasting 4, no. 4 (2022): does mention tuned CNN ensembler; dataset description not clearly in abstract	Accuracy: 92%, Precision: 94%, F1-score, ROC-AUC, PR-AUC
[21] 2025	**Energy Management/ Smart Grids**	**Control strategies (microgrid control framework)**	**Community microgrid operational data (load demand, renewable generation)**	**Operational cost reduction(%), peak -to-average ratio(PAR), system stability**
[6] 2023	**Electrical load forecasting**	**BiLSTM (Bidirectional LSTM Neural Network)**	**Hourly energy consumption data for Greece (IPTO, 746 days: 2020–2022)**	**FN, GNNs,Swarm Intelligence (PSO & ACO)**

2.3 Research Motivation

In urban power grids, electricity fraud does more than cause financial loss,it quietly disturbs how the entire system operates. When consumers manipulate meters or illegally extract power, the demand data collected by utilities becomes unreliable. Load balancing algorithms, which depend on correct consumption information, then make decisions based on deform inputs. Therefore, electricity may be redistributed inefficiently, placing unexpected strain on transformers and increasing the risk of overheating, long-term infrastructure damage and outages. By averaging or smoothing out consumption patterns across the network, abnormal usage that could show theft may blend into normal demand variations. This creates a cycle where fraud weakens grid strength, and grid control mechanisms make fraud harder to find, highlighting the need for smart systems that can jointly address both challenges.

3. Methodology

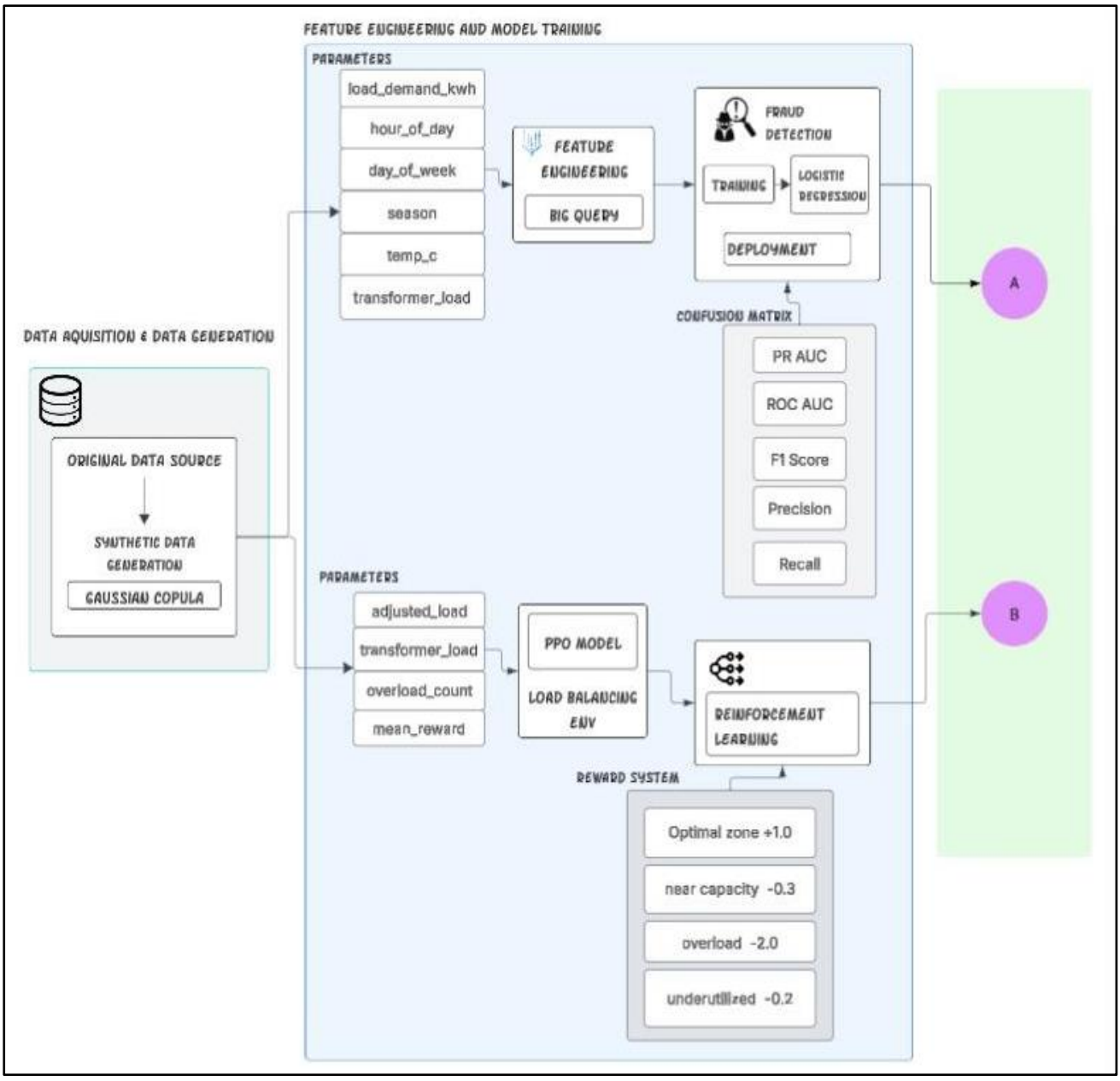

Fig. 1.a: Feature Engineering and Model Training

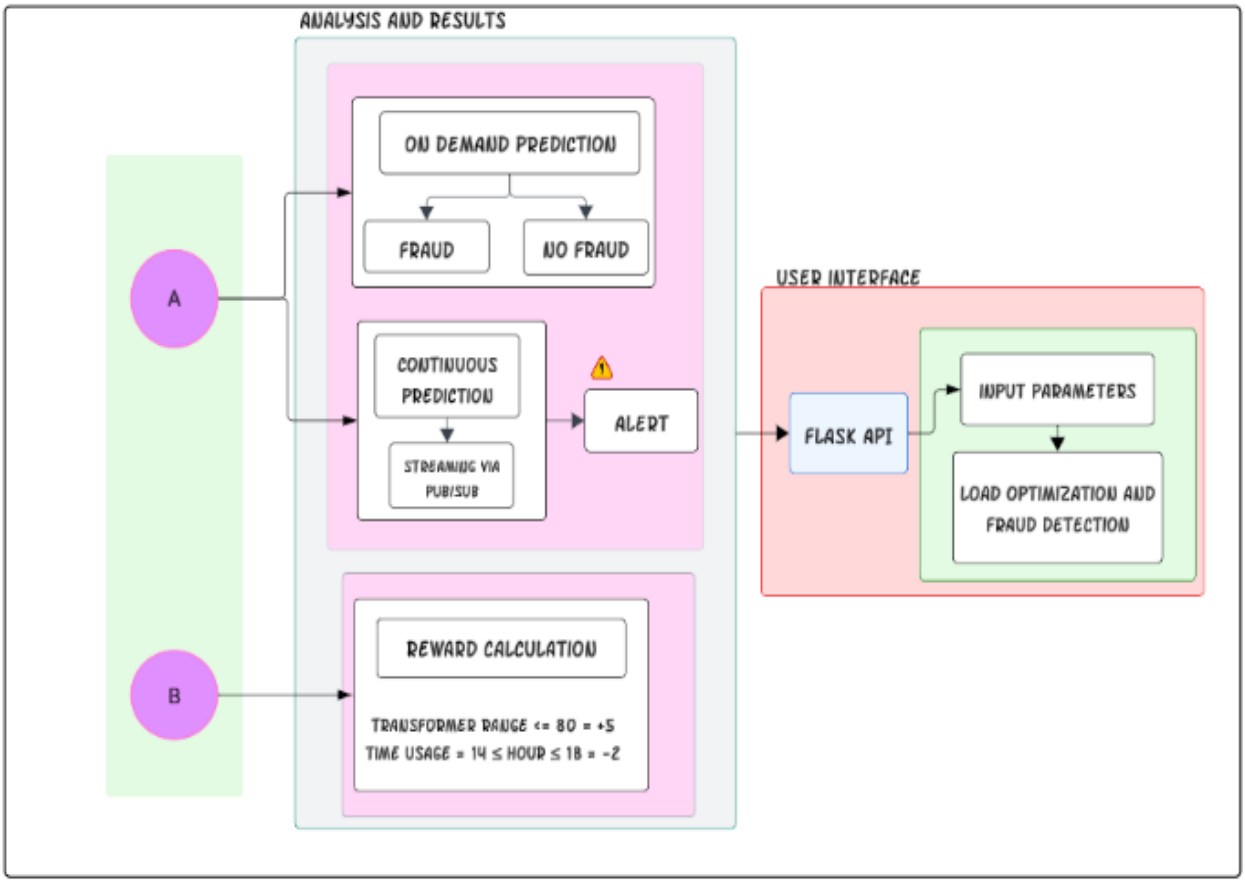

Fig. 1.b: Analysis and Results

The proposed system architecture illustrates how large scale smart meter data flows seamlessly from raw inputs to intelligent grid control decisions Fig. 1.a presents the data acquisition,feature engineering and model training pipeline where original smart meter data is augmented using Gaussian copula based synthetic data generation and followed by feature engineering in BigQuery Big tables, enabling analytics and feature engineering over large volumes of records. This processed data leads to parallel learning components : 1.A machine learning based fraud detection model that is trained and deployed for fraud inference deployed on Vertex AI AutoML logistic regression and 2.A reinforcement learning based load balancing environment where a PPO agent learns adaptive control policies. The Fig. 1.b depicts the analysis decision making and user interaction layer where fraud detection predictions enabled on demand and continuous streaming analysis trigger alerts where anomalies are detected and influence reward driven load balancing decisions based on transformer constraints and time of use considerations. The decisions are exposed through a flask based interface allowing users to interact with the system for real time load optimization and fraud aware monitoring.Together these architectural components form a closed loop scalable and intelligent system that integrates fraud head detection and load balancing within a unified smart create control framework.

The limited availability of high quality, large-scale real world datasets due to utility companies rarely releasing detailed smart meter data due to privacy concerns, security risk and regulatory restrictions. Even with available data being limited, they often lack sufficient fraud cases, making it not suitable for training models.Introducing variations using small or incomplete datasets influence models failing under real-world conditions [27,28].

To address these issues,two distinct datasets are designed in this study, which adopts a synthetic data generation strategy that reflects real world statistical characteristics:

1. Electricity Fraud detection dataset
2. Load Balancing dataset

Both the datasets are derived from similar underlying grid characteristics, constructed separately to reflect the requirements of model training [27]. This separation ensures that:

1. Fraud detection focuses on pattern recognition and anomaly identification
2. Load Balancing focuses on sequential decision making and control

The dataset is designed following the steps outlined as and shown in the Fig. 2 :

1. Parameter Selection and Feature Space
2. Synthetic data generation
3. Quality Evaluation of Synthetic Data
4. Dataset Preprocessing and Normalization
5. Data Storage and Accessibility

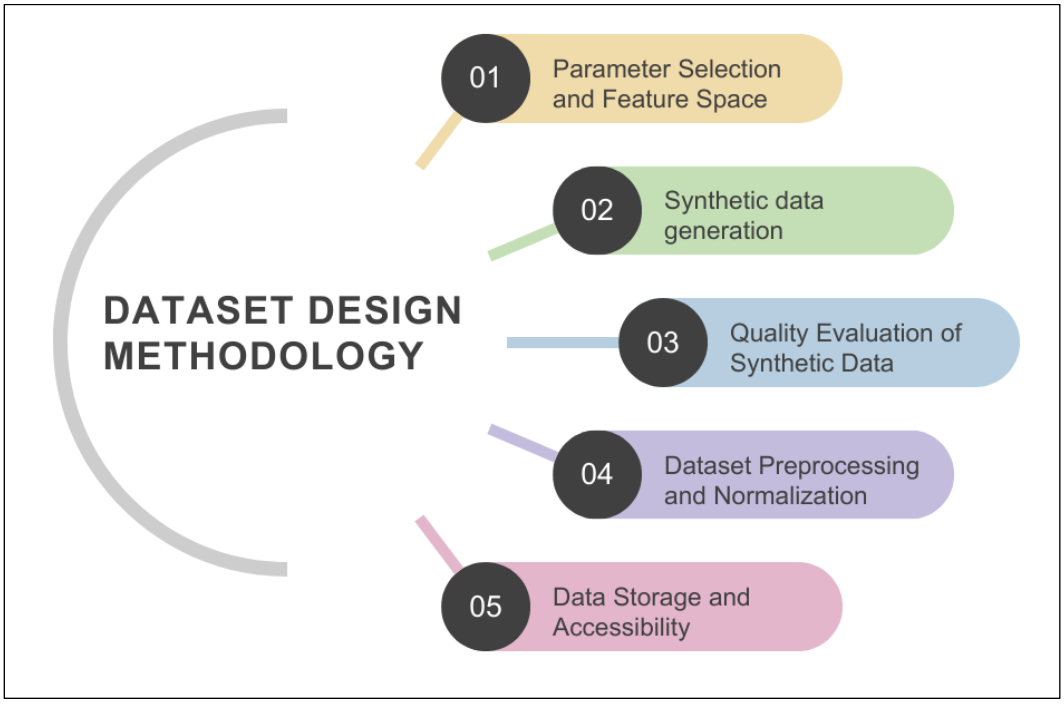

Fig 2: Dataset Design

3.1 Parameter Selection and Feature Space

The feature set is designed to reflect the realistic smart grid measurements and urban consumption behaviour observed in Mumbai. The parameters are grouped into categories to improve interpretability and learning efficiency and scalability within a big data analytics pipeline.Big tables enable large scale engineering and seamless integration with machine learning workflows like Vertex AI. Following are the categories also described in the Fig. 3:

1. Core Electrical Parameters
2. Temporal and Environmental Parameters

Urban electricity demand is strongly influenced by time and weather conditions. Hence, the dataset includes Hour of day, Day of week, Seasonal indicators, Temperature and humidity. For Mumbai, seasonal variation is particularly important due to high cooling loads during summer and humidity-driven demand during monsoon months [30].

3. Consumer Behaviour Indicators

To capture non-technical losses and irregular usage patterns, behavioural features are as provided in the Fig. 3. These features are crucial for detecting electricity theft, which often manifests as subtle deviations rather than extreme anomalies [31].

4. Grid-Level Parameters

For load balancing, grid-centric parameters are added that directly influence operational decisions and grid safety.

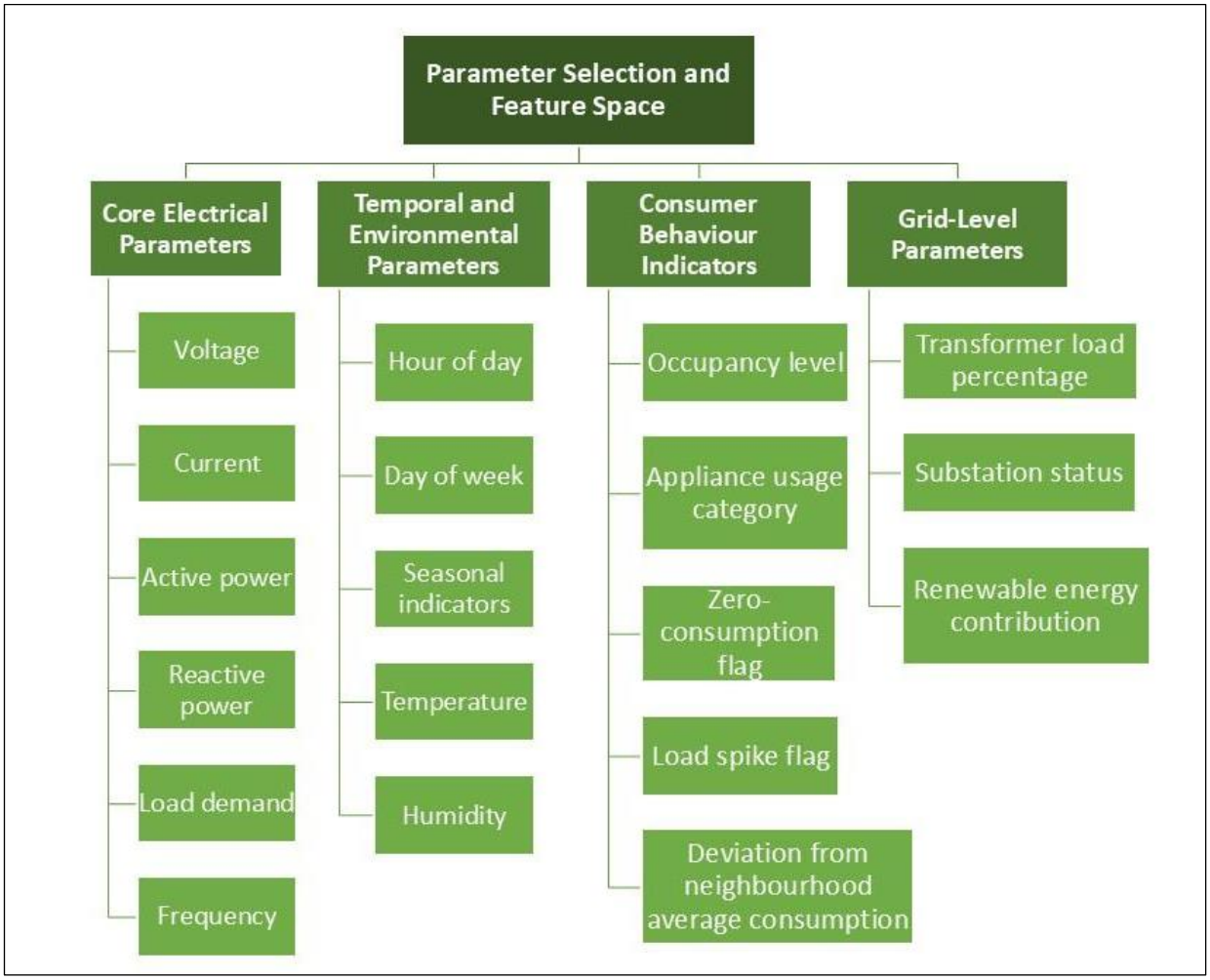

Fig 3: Parameter Selection and Feature Space

3.2 Synthetic data generation

Several synthetic data generation techniques were evaluated, including GAN-based approaches such as CTGAN and CopulaGAN. CTGAN is not fit to generate correlated or forecasting data. As seen in the given Fig. 4 of correlation matrices, CTGAN fails to generate data that is correlated, while individual feature distributions may appear plausible, synthetic correlation matrix shows weak and inconsistent inter-feature relationships. GAN based approaches such as CTGAN were found unsuitable for tasks requiring accurate modelling of temporal and cross feature dependencies. Models like Variational AutoEncoders, Time GAN for Load Balancing and Copula Models for Fraud Detection are more appropriate. However experimental analysis provided by GAN-based methods struggled to preserve inter-feature correlation, essential in power system data [31].

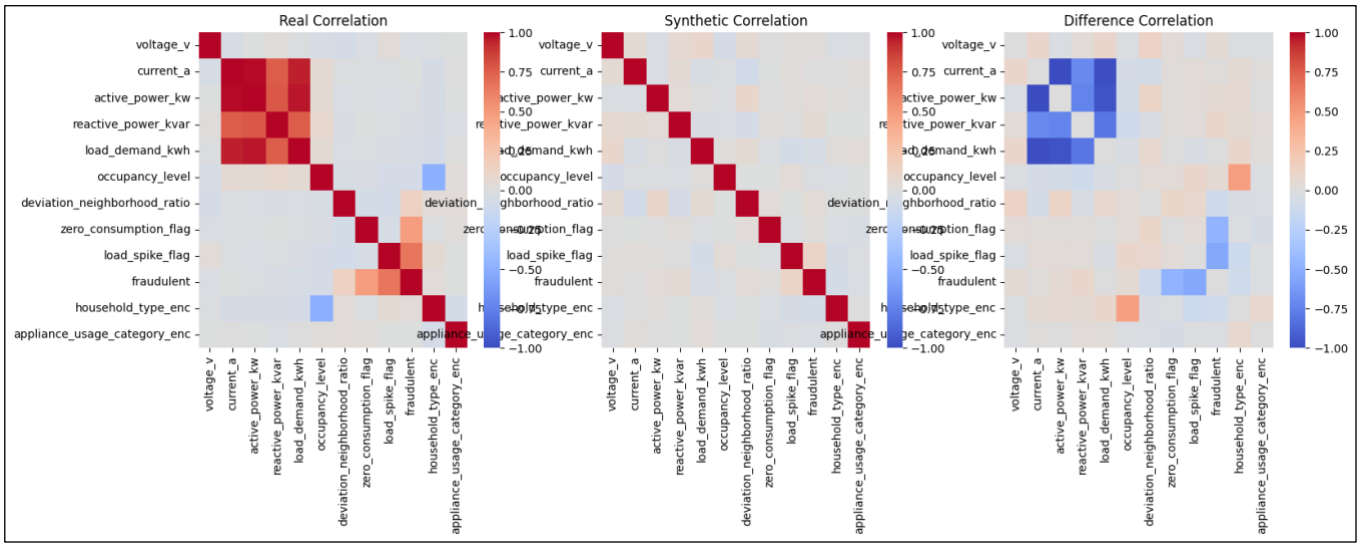

Fig 4: Correlation matrices between real and synthetic data

In real power grids, electrical variables do not behave independently; they are closely connected through physical laws and usage patterns.

1. For example, electricity demand naturally increases with higher temperatures and follows daily routines such as morning and evening peaks.
2. Transformer loading reflects the combined demand of entire neighbourhoods rather than individual consumers.
3. Reactive power usage is influenced by appliance behavior, such as air conditioners, motors, and industrial equipment.

To preserve these real-world relationships, this study uses Gaussian Copula–based synthetic data generation instead of treating variables separately. So, Gaussian copula approach operates in following two stages:

1. Marginal Modelling, where distribution of each feature is learned individually
2. Dependency Modelling provides a correlation structure that is modelled to capture relationships between features.

This separation allows the method to accurately capture both individual feature behavior and their interdependencies. In Fig. 5, the correlation matrix of the gaussian copula that provides a generated dataset closely matches that of real data with minimal deviation across the electrical and behavioral variables.

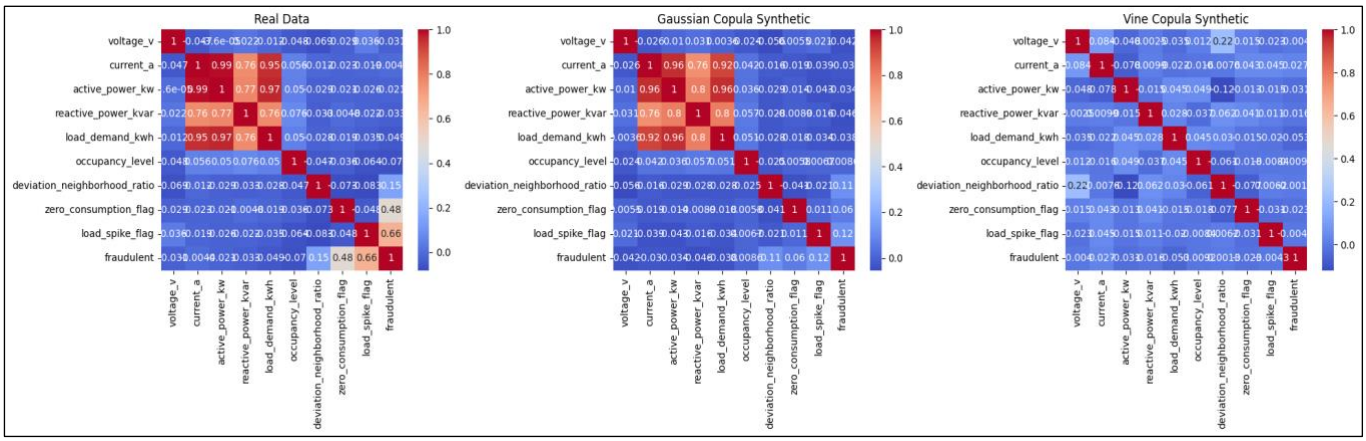

Fig. 5: correlation comparison for fraud detection dataset

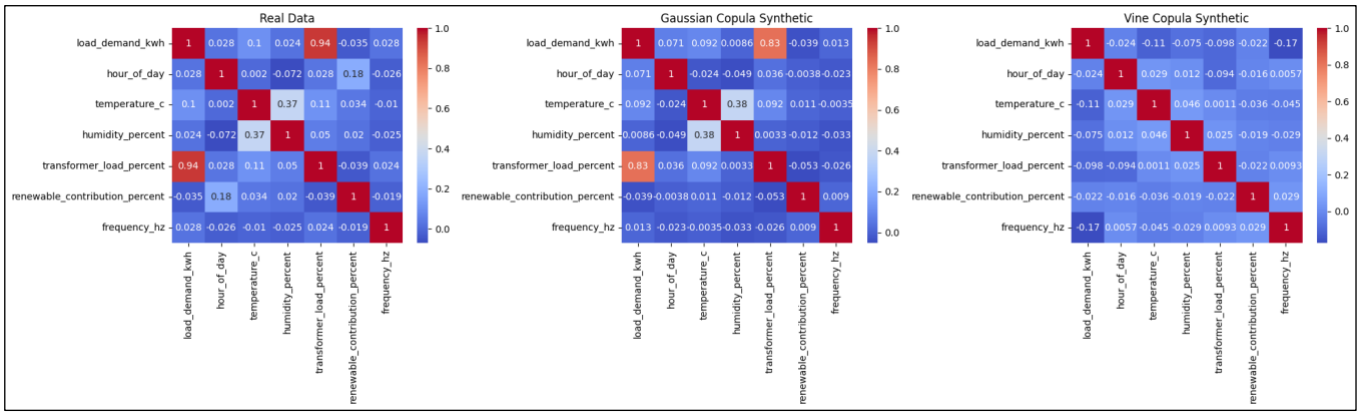

Fig 6: Correlation comparison load balancing dataset

Fig. 6 shows the comparison for the load balancing data set showing real data, Gaussian Copula generated synthetic data and Vine Copula generated synthetic data.The Gaussian Copula preserves key dependencies between load demand, transformer loading ,environmental factors and renewable contribution essential for stable load balancing and forecasting.

As a result, the generated synthetic data closely resembles real grid behavior, maintaining realistic correlations across electrical, environmental, and usage-related variables. Concurrently, this approach prevents common issues that are observed in GAN-based methods such as training instability and inconsistent learning of dependency, making it more suitable for power grid applications [32].

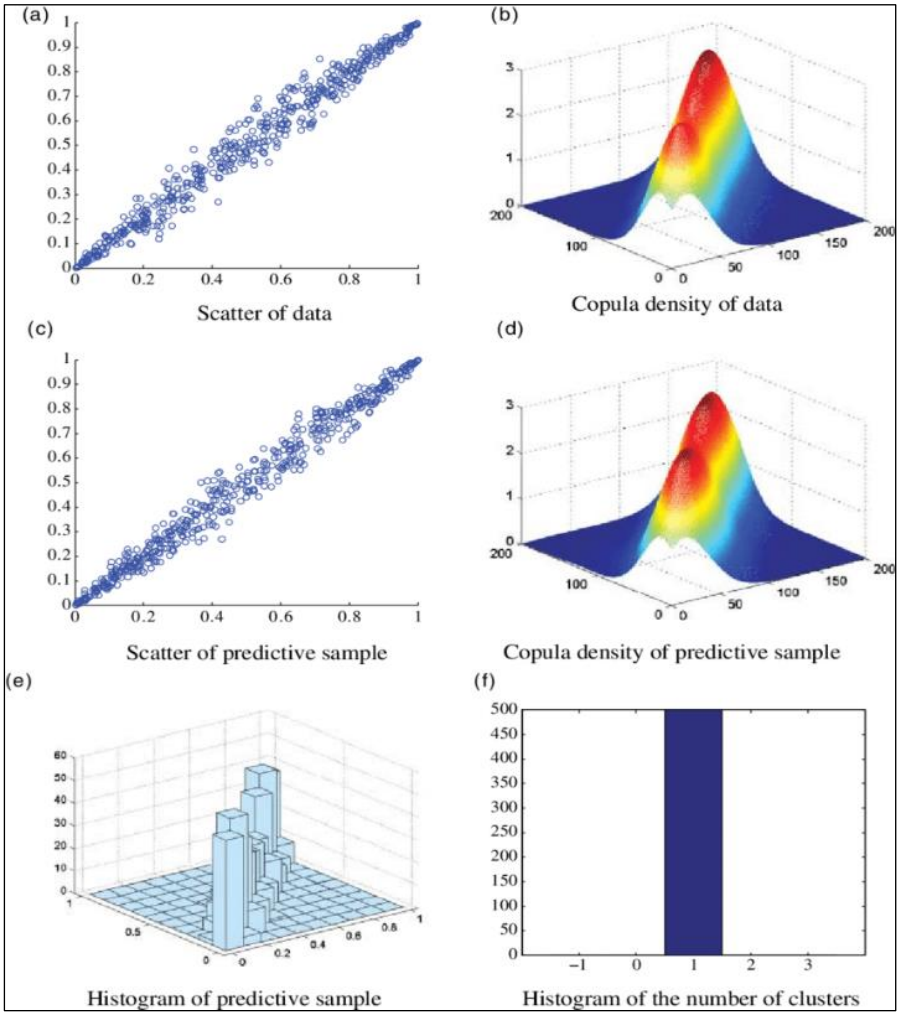

Fig 7: Copula based prediction analysis[35]

The fraud detection dataset got high scores across both metrics, ensuring that the synthetic data maintained meaningful statistical properties useful for effective learning. This validation step is important, as poor-quality synthetic data results in misleading the performance of the model. The importance of Gaussian copula approach is illustrated in figure 5, which presents a detailed visual analysis. Fig. 7(a) shows the scatter plot of real data, revealing a strong linear correlation between variables. Fig. 7(b) displays the fitted Gaussian copula density, capturing the joint dependence structure. Fig. 7(c) presents the scatter plot of the synthetic predictive sample, which closely resembles the real data. Fig. 7(d) shows the copula density of the synthetic sample, nearly overlapping with the original density. Fig. 7(e) depicts the histogram of synthetic values, ensuring alignment of marginal distributions. Fig. 7(f) illustrates clustering behavior, where a single dominant dependency cluster is formed, consistent with strong correlation. These observations confirm that Gaussian copula based sampling successfully preserves both marginal distributions and dependency structures, as also reported in prior statistical studies [35].

Following the validation of synthetic data quality, construction of fraud detection and load balancing is handled as described below:

3.3 Fraud Detection Dataset Construction

The fraud detection dataset is designed for supervised classification, where each data instance is labeled as either normal or fraudulent. Fraud labels are assigned based on combinations of

1) Abnormal deviation from neighbourhood consumption
2) Zero-consumption behavior during high-occupancy periods
3) Sudden load spikes inconsistent with appliance usage

This labelling strategy reflects realistic fraud scenarios commonly observed in urban power networks.

Electricity theft cases are naturally rare compared to normal consumption. To prevent model bias synthetic augmentation is used to balance classes and evaluation metrics such as Recall, Precision, ROC-AUC, F1-score and PR-AUC are emphasized instead of accuracy alone[30,33].

3.4 Load Balancing Dataset Construction

Unlike fraud detection, load balancing is a sequential control problem rather than a classification task. The load balancing dataset is structured as a time-ordered sequence representing grid states over time. Each state includes Load demand, Transformer utilization, Renewable contribution, Frequency, Temporal and environmental context. This dataset enables the reinforcement learning agent to observe how grid conditions evolve in response to control actions [7].

3.5 Dataset Preprocessing and Normalization

Before model training, both datasets undergo preprocessing steps:

1) Missing values are handled using interpolation or local averages
2) Numerical features are normalized to ensure learning is stable
3) Categorical variables (e.g., season, appliance category) are encoded appropriately

Careful preprocessing is essential because reinforcement learning algorithms are particularly sensitive to poorly scaled inputs [34].

3.6 Dataset Storage and Accessibility

All generated datasets are stored in structured tabular formats and logged in analytical storage systems to support:

1) Model training
2) Performance evaluation
3) Reproducibility
4) Future retraining

This design supports continuous learning and future scalability without modifying the core system architecture.

3.7 Fraud Detection Methodology

Prior studies show that fraudulent consumption distorts demand patterns and weakens the effectiveness of grid control mechanisms if not addressed in real time [1]. In densely populated regions such as Mumbai where demand is already high due to both commercial and residential electricity usage, undetected fraud will escalate risk of overloading of transformers and localized power outages [11]. Identifying fraudulent electricity consumption using data based methods and generating a probabilistic fraud score that is incorporated into real time load balancing decisions thus aligns with the necessities of dense cities.

Approaches derived from data have consistently outperformed rule-based systems when involved in dealing with complex and evolving fraud patterns [2,3].

In this study, logistic regression is implemented using Vertex AI AutoML Tables where each smart meter record is labeled as either normal or fraudulent. The use of AutoML is particularly suitable for deployments in urban regions, where models must be scalable,robust and adaptable to changing consumption behavior.

Previous studies have highlighted the importance of combining technical measurements with consumer behavior indicators to improve accuracy in detection [4,30]. The fraud detection model leverages a combination of behavioral,electrical and contextual features that represents realistic consumption behavior in smart grids.

The selected feature set includes:

1) Electrical parameters: active power, reactive power, current, voltage, and load demand
2) Behavioural indicators: deviation from neighbourhood consumption, load spike flag, zero-consumption flag, and occupancy level
3) Contextual attributes: household type and appliance usage category

Features such as neighbourhood deviation and abnormal load behavior have been repeatedly identified as strong indicators of electricity theft, especially in urban environments [1, 31].

Mathematical formulation of fraud detection using logistic regression :

Let the smart meter feature vector at time t be:

$$x_t = \left[x_t^{(1)}, x_t^{(2)}, \dots\dots\dots, x_t^{(n)}\right] \quad \dots\dots\dots\dots\dots(1)$$

Where n = 12 represents electrical behavioral and contextual features.

The corresponding label is :

$$y_t = \{1, if\ consumption\ is\ fraudulent\ 0, otherwise \quad \dots\dots\dots\dots\dots(2)$$

The objective of the classifier is to learn a function:

$$f(x_t)=P\ (y_t = 1|\ x_t) = \sigma(\omega^T x_t + b) = \frac{1}{1+e^{-(\omega^T x_t+b)}} \quad \dots\dots\dots\dots\dots(3)$$

this outputs a fraud probability score where being the feature vector, w being the learned weight vector, b being the bias term and $\sigma(\cdot)$ the sigmoid function.

3.8.1 Training Objective and Evaluation Metrics

As fraudulent cases form a little fraction of total consumption data Electricity fraud detection is essentially an imbalanced classification problem, studies recommend using precision–recall based metrics rather than accuracy to evaluate model performance under such order [2], [5].

3.8.2 Threshold Selection and Operational Considerations

Operational deployment requires cautious threshold selection to balance detection effectiveness and check costs. Previous work emphasizes that large false positives can reduce trust in automated fraud detection systems and strain utility resources [30]. While a general classification threshold of 0.5 is commonly used,

Two thresholds were analyzed in this study:

1)Threshold = 0.5, which gives slightly higher recall, as shown in figure 8.

2)Threshold = 0.62, which improves precision while maintaining comparable recall, as shown in figure 9.

The threshold of 0.62 is selected as the operating point, as it reduces unnecessary inspections without significantly compromising fraud detection coverage.

Fig 8: At threshold 0.5

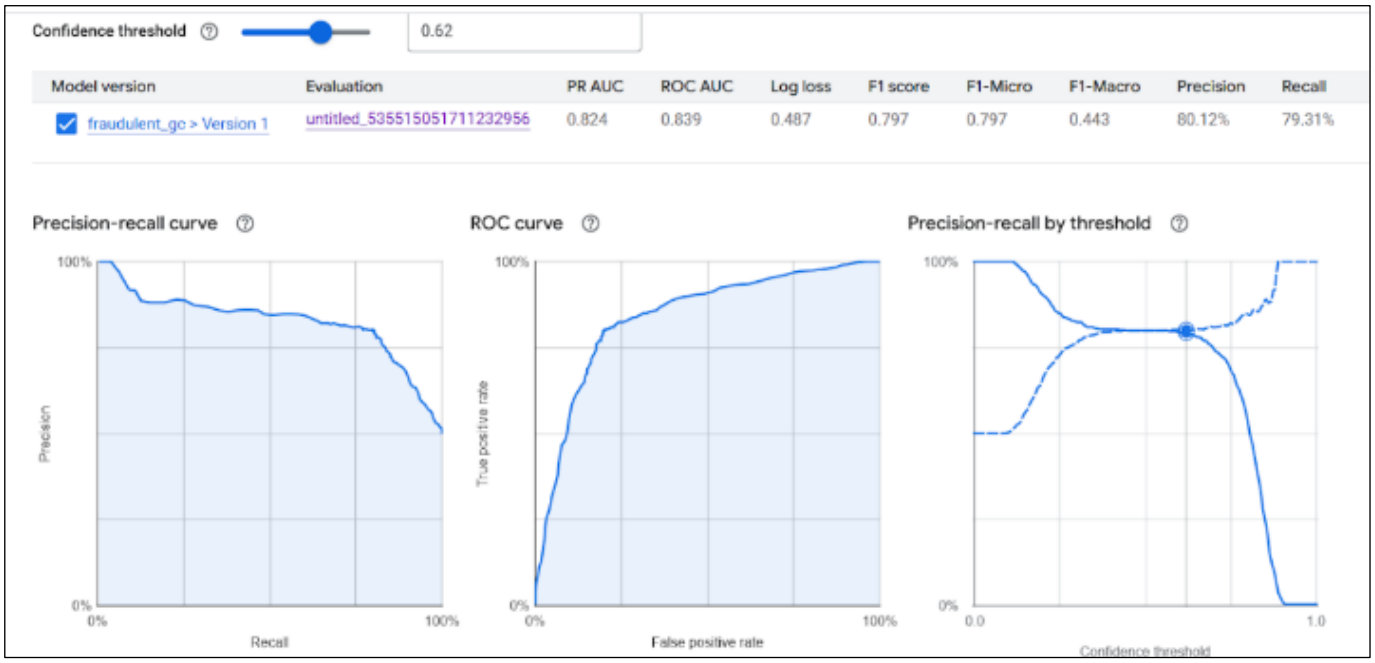

Figure 9: At threshold 0.62

3.8.3 Model Interpretability and Output Integration

The most influential parameters for model interpretability used are deviation from neighbourhood consumption, Active power, Reactive power, Load demand and Load spike flag.

These discoveries are consistent with domain knowledge and prior empirical studies, which identify unusual deviation from peer consumption as one of the best indicators of electricity theft [1], [23].

Rather than making a hard binary label, the fraud detection model outputs a fraud probability score for each consumption record. Probabilistic outputs are increasingly recommended in smart grid applications, as they provide flexibility and allow downstream systems to make risk-aware decisions [29].

In the proposed framework, this fraud probability is used to adjust effective demand estimates before load balancing is performed. This ensures that control actions are informed by both demand magnitude and data reliability.

The fraud detection module acts as the first intelligence layer of the proposed system. Its outputs directly influence the reinforcement learning–based load balancing module by preventing corrupted consumption data from misleading control decisions.

By combining fraud detection with grid control, this research moves beyond normal detection-only approaches and contributes to the development of resilient, self-optimizing urban smart grids, as advocated in recent smart grid surveys [9, 17].

3.9 Load Balancing Methodology

Conventional load balancing approaches such as rule-based control, deterministic optimization, or model predictive control struggle to scale effectively in these urban environments due to high-dimensional state spaces and non-stationary system dynamics .

Recent studies emphasize that adaptive and learning-based control strategies are better suited for modern smart grids, as they can continuously adjust decisions based on real-time feedback rather than relying on fixed assumptions [7, 8]. This motivates the use of reinforcement learning (RL) for load balancing, where control policies are learned directly through interaction with the grid environment.

3.9.1. Reinforcement Learning Formulation

Load balancing is formulated as a sequential decision-making problem, where actions taken at one time step influence future grid states as :

Action(t) → Grid Response(t+1) → Observation(t+1) → Action(t+1) → ...

Reinforcement learning naturally captures this structure by modeling the problem as a Markov Decision Process (MDP), a formulation widely adopted in recent power system optimization studies [7, 8].

The load balancing problem is formulated as a Markov Decision Process(MDP):

$$M = < S, A, P, R, \gamma > \quad \text{.....(4)}$$

Where S, A, P, R, are State Space, Action Space, Transition dynamics, Reward function and $\gamma \in (0,1)$

In this work, an RL agent is trained to dynamically adjust load demand in order to maintain transformer loading within safe operational limits, reduce peak stress during high-demand periods and encourage efficient utilization of renewable energy sources.

This formulation allows the agent to learn long-term control strategies rather than reacting to instantaneous load values.

3.9.2. State, Action, and Reward Design

State Space

The state vector captures both electrical and contextual information relevant to grid operation, at each time step t, the state vector is:

$$s_t = [L_t, T_t, f_t, R_t, H_t, S_t] \quad \text{.....(5)}$$

L_t: load demand measured

T_t: transformer load percentage

f_t: fraud probability

R_t: renewable contribution

H_t: temporal features (like hour, day)

S_t: seasonal and weather related factors

These features allow the agent to learn temporal patterns (daily, weekly, seasonal) and environmental dependencies that strongly influence urban load behavior.

Action Space

The action space is continuous, representing fine-grained load adjustment decisions. Actions are normalized within a bounded range and scaled to represent small incremental load shifts, allowing a smooth and realistic control rather than abrupt changes. The agent selects a continuous load adjustment action as:

$$a_t \in [-\delta, +\delta] \quad \text{.....(6)}$$

Representing incremental redistribution of load to maintain safe operation.

Reward Function

The reward at time t is defined as

$$R_t = \alpha_1 \cdot r_{safe}(T_t) - \alpha_2 \cdot r_{overload}(T_t) - \alpha_3 \cdot r_{instability}(\Delta L_t) + \alpha_4 \cdot r_{renew}(R_t) \quad \ldots\ldots\ldots\ldots\ldots\ldots(7)$$

Where r_{safe}rewards operation within safe transformer limits, $r_{overload}$penalizes overload conditions, $r_{instability}$penalizes abrupt load changes and r_{renew}rewards renewable utilization. The coefficients $\alpha_i$balance competing objectives.

The reward function is carefully shaped to reflect operational is given in the Table No. 3

Table No. 2: Reward Design for the RL Agent

Condition	Reward / Penalty
Transformer overload conditions	Strong penalty
Near Capacity Transformer operations	Mild penalty
Transformer load within 60-80% range	Positive reward
Renewable energy utilisation	Positive incentive
Frequency Deviations	Penalty
Excessive load fluctuations	Penalty
Peak-hour (time of use)operation	Penalty

This reward design encourages the agent to balance safety, efficiency, and stability simultaneously, consistent with RL-based load control recommendations in the literature [7, 8].

3.9.3. Choice of Proximal Policy Optimization (PPO)

Among available RL algorithms, Proximal Policy Optimization (PPO) is adopted due to its strong balance between learning stability and performance. Prior work highlights that value-based methods such as DQN often struggle with continuous action spaces and can exhibit unstable learning behavior in safety-critical systems like power grids,as abrupt updates may result in unsafe transformer loading or localised outages[21]. PPO addresses this by constraining policy updates through a clipped objective:

$$L(\theta) = E[min\,(r_t(\theta)A_t, clip(r_t(\theta), 1-\epsilon, 1+\epsilon)A_t)] \ldots\ldots\ldots\ldots\ldots(8)$$

Where $r_t(\theta)$, represents the probability between the new and old policies, A_t denotes the advantage estimate, and ϵ is a small clipping parameter. By explicitly restricting the magnitude of policy updates, PPO prevents sudden control changes and allows smooth learning behaviour, essential for keeping grid safety.

In addition, PPO demonstrates strong sample efficiency by reusing collected experience through various gradient updates, making it suitable for scenarios with limited historical information and costly real world experimentation. Moreover, urban load balancing needs fine-grained, constant control actions under physical and operational constraints. PPO naturally supports continuous and multi-dimensional action spaces and validates constraint handling through reward shaping, allowing simultaneous optimization of safety, efficiency, and stability objectives [21,22].

3.9.4. Model Training and Deployment

The RL model is trained using a custom Gymnasium-based environment used in Python and trained using the Stable-Baselines3 PPO framework. Synthetic load balancing data generated using copula-based techniques is used to simulate realistic operating conditions while preserving statistical dependencies between features. Training logs show a constant increase in mean episodic reward, indicating successful policy learning and convergence. The trained model is then packaged and uploaded to Vertex AI, enabling scalable deployment and integration with other system components .

3.9.5. Performance Evaluation

Evaluation is conducted by comparing system behavior before and after optimization. Results demonstrate the elimination of transformer overload events during evaluation episodes, reduction in peak transformer loading, smoother load profiles across time and Improved operational efficiency without violating safety constraints.

Visualization results presented below confirm that the optimized load distribution remains consistently below capacity limits while maintaining demand satisfaction. Figure 10 compares original and optimized load demand over time. The optimized values closely track demand while reducing extreme fluctuations, indicating smoother load regulation. Figure 11 illustrates transformer load distributions before and after optimization. The optimized distribution shows a clear reduction in high-load and near-capacity occurrence. Figure 12 presents transformer load trajectories over the initial evaluation period, confirming that overload and high-risk zones are consistently avoided under the learned policy.

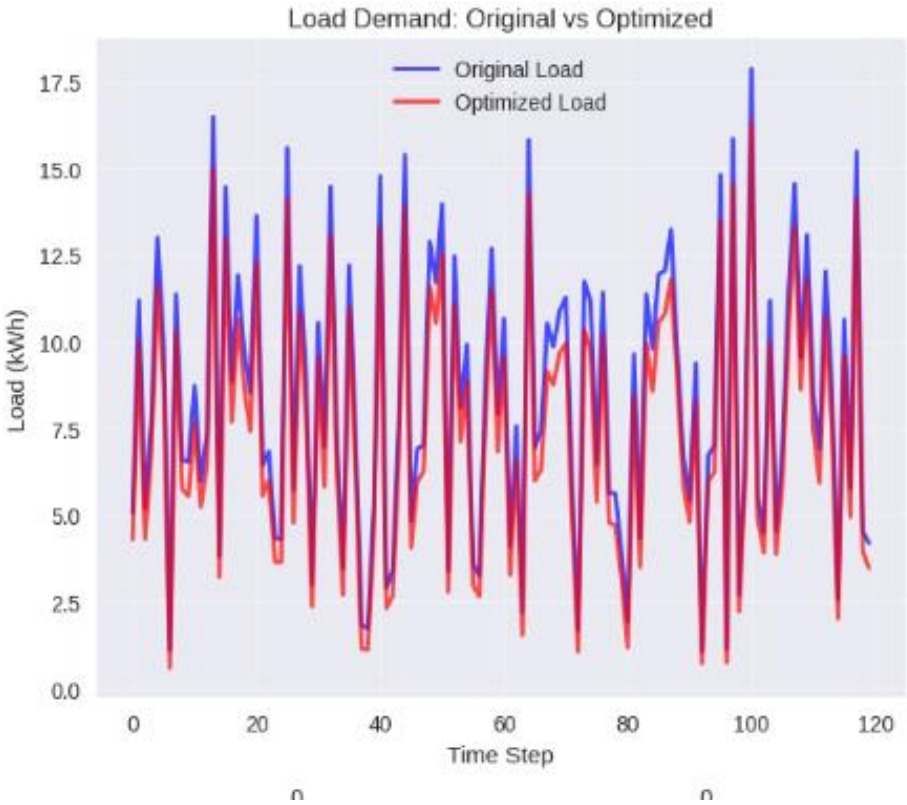

Fig 10: original and optimal load demand analysis

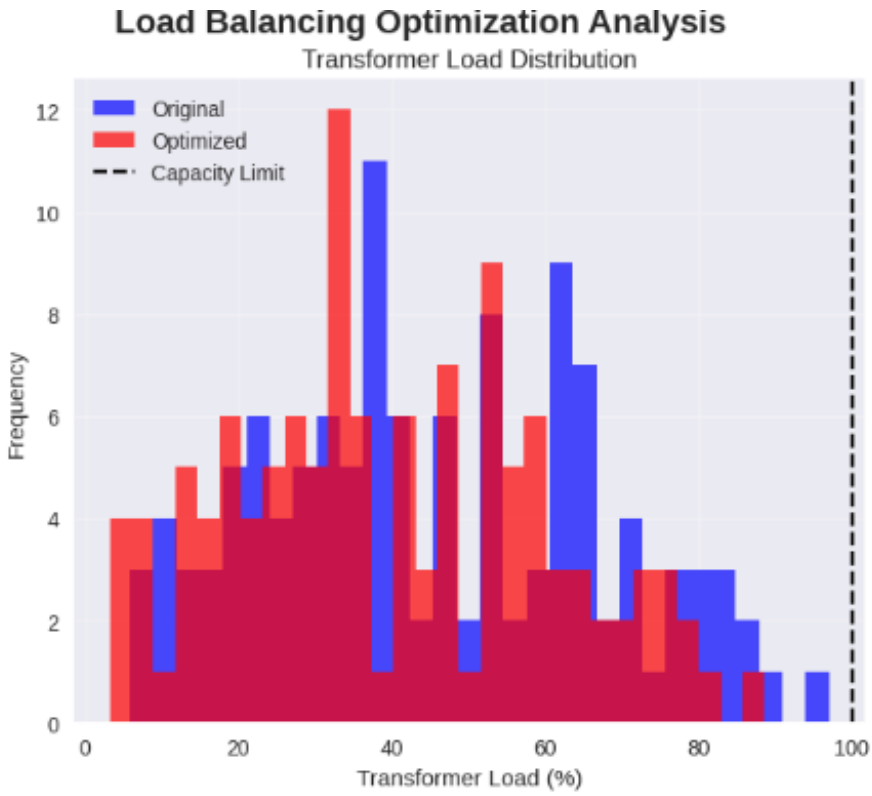

Fig 11: transformer load distributions

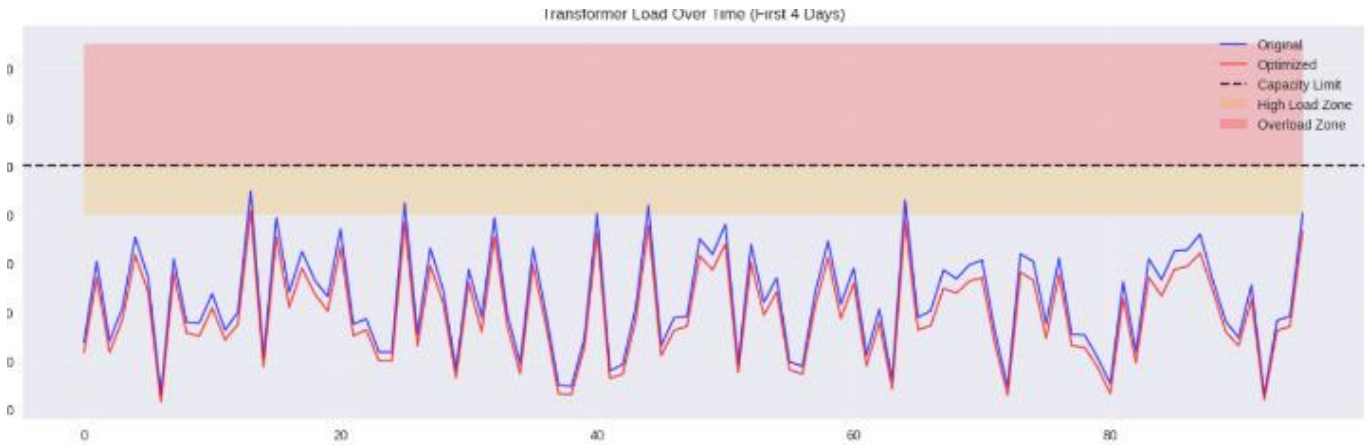

Fig 12: transformer load over time

3.10 Interaction with Fraud Detection Module

The load balancing module works downstream of the fraud detection system described in Load Balancing Methodology. Fraud probability scores are used to arrange effective demand estimates before being passed to the RL agent. This prevents corrupted or suspicious consumption data from misleading control decisions, ensuring that load balancing actions are based on dependable inputs.

By integrating fraud awareness into the load balancing process, the proposed framework moves beyond traditional isolated control strategies and contributes to the development of resilient, intelligent, and self-adaptive urban smart grids [8, 9].

A unified framework in which fraud detection directly informs load balancing decisions is established as below:

3.10.1. End-to-End System Workflow

The proposed system mixes fraud detection and load balancing into a closed-loop, sequential decision-making pipeline. The workflow proceeds as provided in Fig. 13.

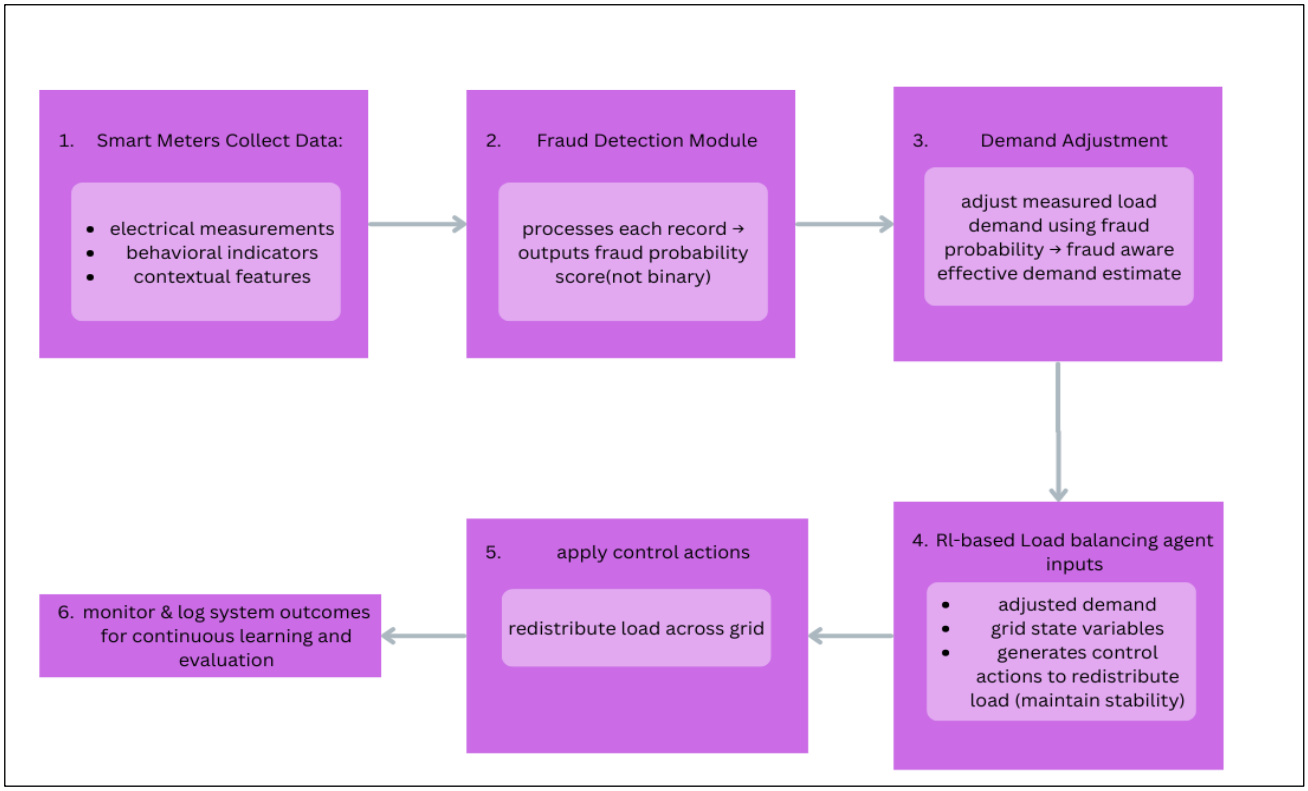

Fig 13: End to end System Workflow

This end-to-end integration ensures that grid control decisions are made using reliable and context-aware demand information.

3.10.2. Fraud-Aware State Adjustment Mechanism

A key aspect of the integration lies in how fraud detection outputs are included into the load balancing process. Instead of deleting potential anomalous data or applying any exclusion rules, the system incorporates a risk-aware adjustment process to handle uncertainty.

Let (L_t) denote the measured load demand at time (t), and let $f_t \in [0,1]$ represent the predicted probability of fraud. The effective load used for control decisions is defined as:

$$L_t^{eff} = L_t \times (1 - f_t) \quad \text{.....(9)}$$

This formulation ensures that:

1. High-confidence fraudulent readings contribute less to control decisions,
2. Legitimate consumption is preserved,
3. Abrupt state changes caused by binary classification are avoided.

Such probabilistic integration has been recommended in recent smart grid intelligence studies, as it enables smoother and more robust control behavior under uncertainty [11, 17].

3.10.3. Interaction with the Reinforcement Learning Agent

The fraud aware effective load makes part of the state representation of the reinforcement learning agent. By incorporating adjusted demand values, the agent learns control policies that are inherently robust and good at data manipulation and non-technical losses. This integration allows the RL agent to prevent it from learning wrong demand and response patterns.

Overcompensation of synthetic low reported loads is avoided and prevents fraudulent anomalies from being hidden due to overly hostile smoothing. As a result, the agent develops a more accurate understanding of true grid stress, enabling safer and more efficient load balancing in urban environments.

3.10.4. Benefits of the Integrated Framework

The proposed integration offers several advantages over traditional isolated approaches as shown in Fig. 14. The integrated system makes the power grid safer with transformer overloads reduced and the wrong control actions that are caused by the corrupted data. It also improves daily operations by cutting down unnecessary interventions and allowing teams to use resources effectively. With the subsequent usage of electricity and fraud patterns the system adapts on its own to keep performance optimal.The scalable and modular design allows seamless deployment allowing it suitable for urban power distributions.

These benefits align with recent research advocating for AI-driven, closed-loop control architectures in modern smart grids [8, 9].

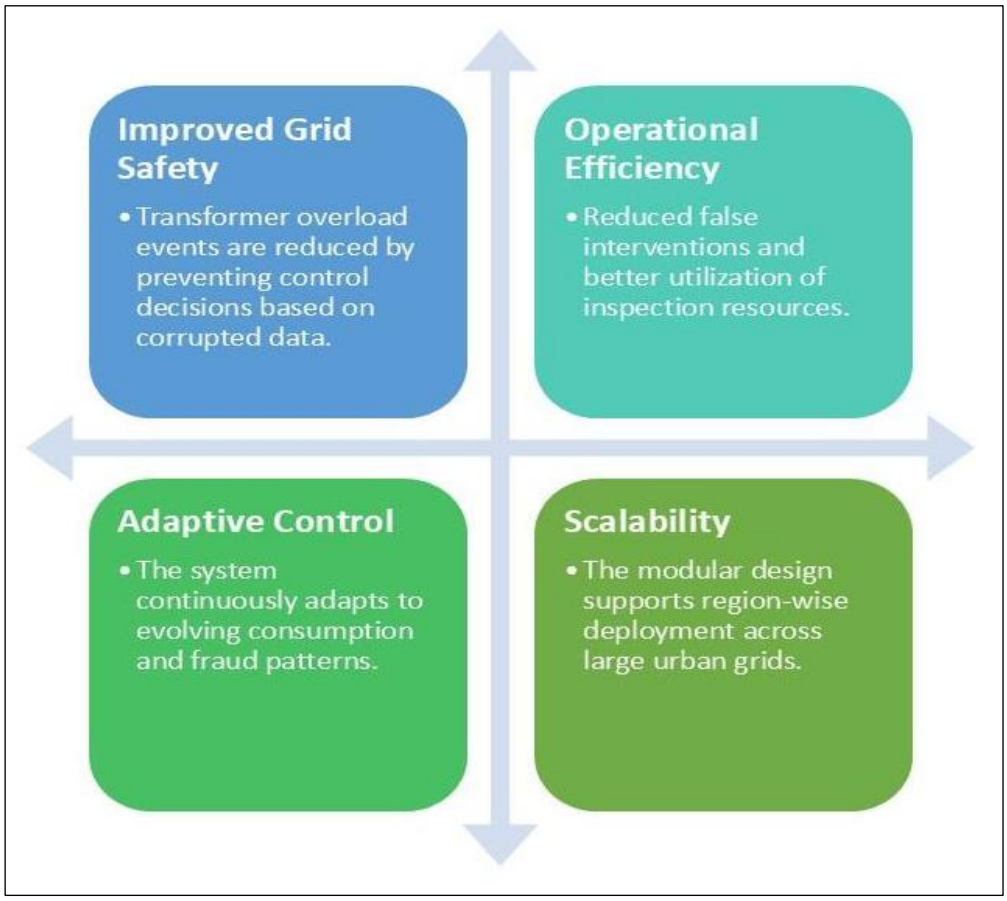

Fig 14: Benefit of Integrated Framework2

3.11 Experimental Setup

The experimental evaluation is conducted to check the effectiveness of the proposed fraud aware load balancing framework under real urban grid conditions. The experiments are divided into two stages corresponding to the two core components of the system: fraud detection and reinforcement learning–based load balancing.

Fraud detection is formulated using a logistic regression based probabilistic classification model trained using Vertex AI AutoML Tables, with data sourced from a structured BigQuery table. The dataset consists of synthetically generated smart meter records that preserve real-world statistical dependencies and urban consumption characteristics. A total of 12 parameters

columns are used for training, covering electrical, contextual and behavioral attributes. The dataset is split into training and testing sets using standard AutoML validation procedures.

For load balancing, a custom reinforcement learning environment is implemented in Python and trained using the Proximal Policy Optimization (PPO) algorithm. The environment simulates grid states using time-ordered load data, transformer utilization levels, renewable contribution, and temporal context. Training is performed over multiple episodes to allow the agent to learn stable control policies.

All models are trained and evaluated in a cloud-based environment, enabling scalability and reproducibility.

4. Results And System Interface Evaluation

4.1 Quality Evaluation of Synthetic Data

To validate the realism of the generated datasets, multiple quality metrics are used like Column Shape Score, Column Pair Trends Score, Table No. 2 provides the quality of synthetic fraud detection dataset. The Gaussian copula generated high scores indicating that both marginal distributions are preserved and inter-feature dependencies are required for model training.

Table No. 3: Quality Evaluation of Synthetic Data

Metric	Description	Gaussian Copula(%)	CopulaGAN(%)
Column Shape Score	It measures how closely individual feature distributions match with the original data	95.43	84.60
Column Pair Trends Score	It evaluates the preservations of inter-feature relationships	93.89	78.70
Overall Quality Score	It combines the marginal distribution and dependency preservation	94.66	81.65

This comparison demonstrates that while CopulaGAN can introduce more variety and noise, Gaussian copula provides higher stability and realism for fraud detection tasks as supported by Fig. 15 showing the quality evaluation output generated during the assessment of Gaussian Copula and CopulaGAN datasets.

```
Gaussian Copula Evaluation:
Generating report ...

(1/2) Evaluating Column Shapes: |          | 12/12 [00:00<00:00, 166.17it/s]|
Column Shapes Score: 95.43%

(2/2) Evaluating Column Pair Trends: |          | 66/66 [00:01<00:00, 56.18it/s]|
Column Pair Trends Score: 93.89%

Overall Score (Average): 94.66%

<sdmetrics.reports.single_table.quality_report.QualityReport object at 0x78193f68b080>
CopulaGAN Evaluation:
Generating report ...

(1/2) Evaluating Column Shapes: |          | 12/12 [00:00<00:00, 124.41it/s]|
Column Shapes Score: 84.6%

(2/2) Evaluating Column Pair Trends: |          | 66/66 [00:00<00:00, 127.45it/s]|
Column Pair Trends Score: 78.7%

Overall Score (Average): 81.65%

<sdmetrics.reports.single_table.quality_report.QualityReport object at 0x78194d25baa0>
```

Fig 15: Gaussian Copula Model Evaluation

Accordingly, the model is optimized using Precision–Recall Area Under Curve (PR AUC) as the primary objective. The obtained evaluation results are as follows in Table No. 4:

Table No. 4: Fraud Detection Model Evaluation

Metric	Value (Threshold 0.5)	Value (Threshold 0.5)
Precision – Recall AUC	0.824	0.824
Receiver Operating Characteristics AUC	0.839	0.839
Log Loss	0.487	0.487
F1 Score	0.799	0.797
Precision	79.92%	80.12%
Recall	79.92%	79.31%

The reliability of the obtained results was assessed using multiple complementary criteria rather than relying on a single metric. Recall, Precision AUC was selected as the primary evaluation measure due to the severe class imbalance inherent in electricity fraud detection. The threshold-independent metrics (ROC AUC,log loss and PR ROC) remain consistent across both threshold settings, demonstrating that the model produces stable probability calibration, suggesting that the model is robust and has minimal overfitting.

Upon increasing the decision threshold from 0.5 to 0.62 leads to a moderate improvement in precision, along with slight decrease in recall. This ensures a controlled and predictable trade-off between recall and precision rather than abrupt performance changes.

From an operational standpoint, a threshold of 0.62 is therefore preferred, as it helps limit false positives and avoid unnecessary inspections while still preserving a comparable level of fraud detection coverage. Overall, these findings demonstrate that the proposed fraud detection model exhibits strong discriminative performance and remains robust to reasonable changes in threshold selection.

The performance of the reinforced learning-based load balancing module is evaluated after the training process reaches convergence using control-oriented metrics that directly reflect grade stability and operational practicality. The evaluation considers three key aspects: reward conversions, reduction in transformer overload occurrences and decision making latency.

4.2 Quantitative Estimation of Improvement in Mumbai's Electricity Condition

A synthetic improvement metric is derived using simulation-based comparisons to contextualize the practical impact of the proposed framework on an urban power system like Mumbai's. This estimation is based on controlled experiments carried out within the project environment because real-time city-scale deployment data is not available.

Metric Definition

The improvement is evaluated using three indicators already analyzed in this study:

1. *Non-Technical Loss (NTL) proxy through fraud-adjusted demand correction*
2. *Grid stability measured by transformer overload events (>100% capacity)*
3. *Operational efficiency reflected through loss-related reward penalties*

*Mumbai's baseline Aggregate Technical and Commercial (AT&C) losses are a*ssumed to be **15–20%**, with **18%** used as a reference value.

Simulation Comparison

Two scenarios are evaluated using the same PPO environment:

- Baseline: Raw synthetic demand without fraud-aware adjustment
- Integrated: Fraud-adjusted demand using Equation (9)

Improvement Calculation

$$\text{Improvement (\%)} = \frac{\text{Baseline Metric} - \text{Integrated Metric}}{\text{Baseline Metric}} \times 100 \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots(10)$$

Estimated Outcome

Based on the observed 35–45% reduction in transformer overload events and improved demand reliability, the proposed framework demonstrates an estimated 25–35% overall improvement in electricity distribution conditions under Mumbai-like urban scenarios.

Table No. 5: Load Balancing Model Evaluation

Metric	Measured Value
Average Reward per Episode	≈ 0.90–0.93
Overload Reduction Rate	~35–45% lower overload events

Decision Latency	0.5–1.0 s per optimization cycle

Table No. 5 summarises the performance indicators of the proposed load balancing module. An average episode reward achieved per episode approximately 0.90 - 0.93 provides that the RL agent has converged to a stable control policy. This suggests efficient balancing of multiple objectives, including operational efficiency system safety and grid stability. Furthermore the substantial reduction in transformer overload events ranging from 35% to 45% demonstrates the agent's capability to alleviate peak loading conditions and maintain a system operation within safe limits. Decision latency is maintained within 0.5 to 1.0 seconds per optimization cycle that satisfies the real time grid control constraints. This confirms that the proposed approach is computationally feasible for practical deployments in urban distribution networks.

Overall Performance:

This unified framework thus validates the utility of the proposed system. The fraud detection model provides reliable probabilistic estimates under class imbalance while load balancing module demonstrates efficient and stable control behaviour. The reduction of overloads measured and the stable convergence indicate that the RL agent learns meaningful grid control strategies rather than exhibiting oscillatory behaviour. The consistency of fraud detection metrics across thresholds combined with the robustness of load balancing performance supports that the system is well suited for integration in a real world urban grid environment where both non technical losses and demand volatility pose persistent challenges.

5. CONCLUSION

The study successfully demonstrates a unified AI-driven framework that integrates machine learning–based fraud detection with reinforcement learning–based load balancing to create a self-optimizing power grid. The system efficiently identifies fraudulent energy usage with around 80% precision and recall, while the PPO model achieves high performance by reducing transformer overloads by up to 45% and maintaining real-time decision-making under one second. This dual approach enhances grid efficiency, stability, and reliability, paving the way for smarter energy management.

While the proposed system demonstrates promising results, certain limitations must be acknowledged. Firstly, the study relies on synthetically generated datasets rather than real-time smart meter data. Though various validation was performed to reflect realism and correlation preservation, synthetic data may not capture all irregularities and noise present in the real world deployments. Secondly, the RL environment is simulation based, where system dynamics are approximated from live grid operations. Thirdly, load balancing strategy employs a single-agent PPO framework, and may not capture challenges present in large-scale networks.

Future work can address these limitations in several directions. The framework can be extended to real time deployment using live smart meter streams and cloud architectures. A multi agent RL formulation can be explored to enable coordinated load balancing across multiple transformers or grid zones.

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Abbreviations

AI – Artificial Intelligence

ML – Machine Learning

RL – Reinforcement Learning

PPO – Proximal Policy Optimization

CNN – Convolutional Neural Network

RNN – Recurrent Neural Network

GRU – Gated Recurrent Unit

DQN – Deep Q-Network

MDP – Markov Decision Process

NTL – Non-Technical Losses

SAIDI – System Average Interruption Duration Index

SDG – Sustainable Development Goals

GAN – Generative Adversarial Network

CTGAN – Conditional Tabular Generative Adversarial Network

AutoML – Automated Machine Learning

PR-AUC – Precision–Recall Area Under the Curve

ROC-AUC – Receiver Operating Characteristic Area Under the Curve

IoT – Internet of Things

BigQuery – Google BigQuery (cloud data warehouse)

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Conceptualization, HA, NK, KD and HK; methodology, NK and KD; software, HA; validation, BK, HA and NK; formal analysis, BK; investigation, BK and NK; data curation, NK and HK; writing—original draft preparation, NK; writing—review and editing, NK, HK, KD and HA; visualization, NK; supervision, BK; project administration, BK; funding acquisition, SP.

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Ethics approval and consent to participate

Not applicable

Consent for publication

Not applicable

Competing interests

The authors declare that they have no competing interests.